

Homework 1: Corporate Finance, Ross et al., 11e

For GBF COF 2026 Autumn

Please upload your answers and solutions to **Elearning: 'Homework1'**

Please name your **PDF file** as "**Name-Student ID-Solution 1**"

Please submit the homework by **October 11, 2026**

Conceptual and factual

(The answer to each question **should not exceed 100 words**)

1. **A simplified explanation of what the Perpetuity/Growing perpetuity/annuity/Growing annuity means.**
2. **What are the three common forms of business organizations? State three key features of "Corporation".**
3. **Interest Rates: What happens to the future value of an annuity if you increase the rate r ? What happens to the present value?**
4. **Goal of the Firm: Evaluate the following statement: Managers should not focus on the current stock value because doing so will lead to an overemphasis on short-term profits at the expense of long-term profits.**

Quantitative

1. Chapter 2, Problem Set 5, Calculating OCF

Barrett, Inc., has sales of \$19,800, costs of \$10,900, depreciation expense of \$2,100, and interest expense of \$1,250. If the tax rate is 40 percent, what is the operating cash flow, or OCF?

2. Chapter 4, Problem Set 3/4/5, Calculating Present Values/Interest Rates/Number of Periods

Present Value \$	Years	Interest Rate %	Future Value \$
	8	7	13,827
	17	11	725,380
242	4		345
410	8		927
625		7	1,284
810		12	4,341

3. Chapter 5, Problem Set 10, Problems with IRR

Suppose you are offered \$9,400 today but must take the following payments:

Year	Cash Flows \$
0	9,400
1	-4,500
2	-3,100
3	-2,400
4	-1,800

- What is the IRR of this offer?
- If the appropriate discount rate is 10 percent, should you accept this offer?
- If the appropriate discount rate is 20 percent, should you accept this offer?
- What is the NPV of the offer if the appropriate discount rate is 10 percent?
20 percent?
- Are the decisions under the NPV rule in part (d) consistent with those of the IRR rule?

4. Chapter 6, Problem Set 3, Calculating Project NPV

Down Under Boomerang, Inc., is considering a new three-year expansion project that requires an initial fixed asset investment of \$1.65 million. The fixed asset will be depreciated straight-line to zero over its three-year tax life, after which it will be worthless. The project is estimated to generate \$1.24 million in annual sales, with costs of \$485,000. The tax rate is 35 percent and the required return is 12 percent. What is the project's NPV?

5. Chapter 7, Problem Set1, Sensitivity Analysis and Break-Even Point

We are evaluating a project that costs \$588,000, has an eight-year life, and has no salvage value. Assume that depreciation is straight-line to zero over the

life of the project. Sales are projected at 70,000 units per year. Price per unit is \$36, variable cost per unit is \$20, and fixed costs are \$695,000 per year. The tax rate is 35 percent, and we require a return of 15 percent on this project.

- a. Calculate the accounting break-even point.
- b. Calculate the base-case cash flow and NPV. What is the sensitivity of NPV to changes in the sales figure? Explain what your answer tells you about a 500-unit decrease in projected sales.
- c. What is the sensitivity of OCF to changes in the variable cost figure?